

Leen-Noemi Alexandre
Manglani
Dealing with Data
2 March 2026

Operational and Financial Drivers of Hospital Operating Margin

This project uses the **2023 Medicare Hospital Cost Report dataset** ([NEW CostReport_2023_Final.csv](#)). The dataset contains hospital-level financial, operational, and structural variables for thousands of hospitals across the United States. While the data was originally provided as a single CSV file, the dataset was imported into PostgreSQL.

The dataset was normalized into four separate relational tables (hospitals, financials, utilization) to eliminate redundancy, preserve referential integrity under 3NF principles, and reflect the logical separation of structural, financial, operational, and cost components. All tables are connected through a shared hospital identifier ([provider_ccn](#)).

The first and most important table is the **hospitals, Hospital Characteristics Table**. The table has the columns and rows focusing on Hospital name and location (city, state, county), rural_versus_urban, ccn_facility_type, provider_type, type_of_control (ownership/control classification), and number_of_beds. I decided that the primary key would be: [provider_ccn](#).

The second table is **financials (Financial Outcomes Table)**. This table provides the financial outcome measures used to compute operating margin, which is what my problem statement is based on. The foreign key is [provider_ccn](#). This table contains revenue and income variables, including net_patient_revenue, net_income, total_income, and total_patient_revenue.

The third table is **costs (Cost Structure Table)**. This table captures the cost allocation and uncompensated care burden of the hospitals. The foreign key is [provider_ccn](#). This table contains expense and cost burden variables, including total_costs, total_salaries_adjusted, total_unreimbursed_and_uncompensated_care, cost_of_charity_care, total_bad_debt_expense, and depreciation_cost.

The fourth table is **utilization (Operational Activity Table)**. This table measures service intensity and operational scale. The primary key *and* foreign key is [provider_ccn](#). This table contains utilization metrics, including fte_employees_on_payroll, total_days_all_payers, total_bed_days_available, and total_discharges_all_payers.

This project examines which hospital characteristics and operational factors are associated with hospital financial efficiency, measured by operating margin. Operating margin is defined as: $\text{Operating Margin} = \text{Net Income} / \text{Net Patient Revenue}$. Operating margin reflects how much profit a hospital retains for each dollar of revenue generated. By connecting hospital characteristics, utilization metrics, and cost structures across normalized tables, the analysis aims to identify which factors are most strongly related to higher operating margins and overall financial efficiency. The central research question is: Can Short-Term Hospitals (STH) operating margin be predicted using ownership type, service utilization (occupancy rate), and cost structure variables (labor cost ratio and uncompensated care burden)?

I chose this dataset because I have a strong interest in the intersection between healthcare and business, particularly how financial incentives shape hospital operations. In a class last semester, we discussed the growing role of private equity in healthcare and its effects on hospitals, including changes in staffing decisions and financial priorities. That discussion highlighted the importance of understanding hospital finances not just from a clinical perspective, but as business entities operating under complex economic pressures.

The Hospital Provider Cost Report dataset provides a concrete way to explore these issues using real-world data. By linking hospital characteristics and utilization metrics to cost and financial performance, the dataset allows for analysis of how operational decisions translate into financial outcomes. This aligns closely with my interest in examining how business models and financial strategies influence healthcare delivery, making the dataset and problem statement both personally meaningful and analytically well-suited to the goals of this project.

The analytic dataset was constructed by integrating four normalized tables joined using `provider_ccn` as the relational key. This relational structure ensured that hospital characteristics, financial performance, operational utilization, and cost structure could be analyzed within a unified framework.

The initial operating margin calculations using: $\text{net_income} \div \text{net_patient_revenue}$ produced extreme and economically implausible values, including margins below $-100,000,000\%$. I investigated and noticed that several hospitals reported net patient revenue values equal to 1 or other extremely small denominators, creating explosive ratio distortions. To prevent this, the revenue floor of $\text{net_patient_revenue} > 1,000,000$ was imposed. This restriction ensures that only economically meaningful hospital-scale institutions are included in the analysis.

Even after imposing the revenue floor, extreme ratio outliers remained. To remove them while preserving meaningful financial variation, the operating margin was restricted to BETWEEN -1 AND 1. This retains substantial variability while eliminating distortion from reporting irregularities. The final clean analytic dataset is restricted to Short-Term Hospitals (STH) to maintain comparability; requires positive denominators for all constructed ratios; removes NULL financial values; excludes extreme operating margin outliers; applies a revenue

floor to prevent small-denominator distortions. Thus, after cleaning, the final analytic sample includes 3,103 hospitals.

After constructing the analytic variables, I examined the distributions of operating margin and the key cost ratios. Operating margins display heavy tails, with a small number of extreme observations. Because linear regression can be sensitive to outliers, the operating margin was restricted to the interval $-1, 1$ to remove implausible values while preserving meaningful variation in hospital financial performance.

To ensure economically interpretable analysis, performance measures were constructed directly in SQL using **CAST** operations for numeric precision:

- **Operating Margin** = Net Income ÷ Net Patient Revenue
- **Occupancy Rate** = Total Days All Payers ÷ Total Bed Days Available
- **Labor Cost Ratio** = Total Salaries Adjusted ÷ Total Costs
- **Uncompensated Care Ratio** = Total Unreimbursed and Uncompensated Care ÷ Total Costs

These constructed ratios allow evaluation of financial efficiency in relation to ownership structure, utilization intensity, labor allocation, and uncompensated care burden. The SQL query used to generate the final analytic dataset is shown below:

```
SELECT
  h.provider_ccn,
  h.hospital_name,
  h.state_code,
  h.rural_versus_urban,
  h.type_of_control,
  h.number_of_beds,

  f.net_income,
  f.net_patient_revenue,
  (CAST(f.net_income AS NUMERIC) /
   CAST(f.net_patient_revenue AS NUMERIC)) AS operating_margin,

  u.total_days_all_payers,
  u.total_bed_days_available,
  (CAST(u.total_days_all_payers AS NUMERIC) /
   CAST(u.total_bed_days_available AS NUMERIC)) AS occupancy_rate,

  c.total_costs,
  c.total_salaries_adjusted,
  (CAST(c.total_salaries_adjusted AS NUMERIC) /
   CAST(c.total_costs AS NUMERIC)) AS labor_cost_ratio,
```

```

        c.total_unreimbursed_and_uncompensated_care,
        (CAST(c.total_unreimbursed_and_uncompensated_care AS NUMERIC) /
        CAST(c.total_costs AS NUMERIC)) AS uncompensated_ratio

FROM hospitals h
JOIN financials f
    ON h.provider_ccn = f.provider_ccn
JOIN utilization u
    ON h.provider_ccn = u.provider_ccn
JOIN costs c
    ON h.provider_ccn = c.provider_ccn

WHERE
    h.ccn_facility_type = 'STH'
    AND CAST(f.net_patient_revenue AS NUMERIC) > 1000000
    AND f.net_income IS NOT NULL
    AND CAST(c.total_costs AS NUMERIC) > 0
    AND CAST(u.total_bed_days_available AS NUMERIC) > 0
    AND (CAST(f.net_income AS NUMERIC) /
        CAST(f.net_patient_revenue AS NUMERIC)) BETWEEN -1 AND 1;

```

EDA Query 1 — Operating Margin Distribution

```

SELECT
    MIN(CAST(net_income AS NUMERIC) / CAST(net_patient_revenue AS NUMERIC)) AS
min_margin,
    MAX(CAST(net_income AS NUMERIC) / CAST(net_patient_revenue AS NUMERIC)) AS
max_margin,
    AVG(CAST(net_income AS NUMERIC) / CAST(net_patient_revenue AS NUMERIC)) AS
avg_margin
FROM hospitals h
JOIN financials f
    ON h.provider_ccn = f.provider_ccn
WHERE
    h.ccn_facility_type = 'STH'
    AND CAST(f.net_patient_revenue AS NUMERIC) > 1000000
    AND f.net_income IS NOT NULL
    AND (CAST(f.net_income AS NUMERIC) /
        CAST(f.net_patient_revenue AS NUMERIC)) BETWEEN -1 AND 1;

```

This query evaluates the distribution of operating margins for short-term hospitals (STH) with revenue greater than \$1 million, excluding mechanically extreme values beyond $\pm 100\%$. Results indicate a minimum operating margin of -97.9% , a maximum of 99.5% , and a mean of 4.3% .

A margin of -97.9% implies that a hospital is losing nearly \$0.98 for every \$1.00 of patient revenue, while a margin of 99.5% indicates a hospital retaining nearly all revenue as net income. Although these extremes are rare, they illustrate the substantial financial heterogeneity present across institutions.

To better assess distribution shape, percentile analysis was conducted. The 10th percentile operating margin is -13.0%, the 25th percentile is -3.3%, the median (50th percentile) is 4.55%, the 75th percentile is 13.0%, and the 90th percentile is 23.0%. The interquartile range (-3.3% to 13.0%) indicates that the central 50% of hospitals operate within a moderate and economically plausible margin band. While the tails extend outward, extreme performance is concentrated within a relatively small subset of facilities.

Importantly, the median operating margin (4.55%) is nearly identical to the mean (4.3%), suggesting limited skewness and indicating that extreme observations do not materially distort the average. This close alignment supports the use of the mean as a representative measure of central tendency for the sample.

Overall, the distribution exhibits moderate dispersion but no pathological clustering in the tails. The average margin of 4.3% aligns with industry benchmarks, which typically place sustainable hospital net margins between 2% and 5%, reinforcing the economic plausibility of the dataset.

EDA Query 2 — Ownership vs Margin

```
SELECT
  h.type_of_control,
  COUNT(*) AS n_hospitals,
  AVG(CAST(f.net_income AS NUMERIC) /
    CAST(f.net_patient_revenue AS NUMERIC)) AS avg_operating_margin
FROM hospitals h
JOIN financials f
  ON h.provider_ccn = f.provider_ccn
WHERE
  h.ccn_facility_type = 'STH'
  AND CAST(f.net_patient_revenue AS NUMERIC) > 1000000
  AND f.net_income IS NOT NULL
  AND (CAST(f.net_income AS NUMERIC) /
    CAST(f.net_patient_revenue AS NUMERIC)) BETWEEN -1 AND 1
GROUP BY h.type_of_control
ORDER BY avg_operating_margin DESC;
```

This analysis examines variation in operating margins across ownership types for short-term hospitals (STH) with net patient revenue greater than \$1 million. Ownership type (type_of_control) classifies hospitals according to institutional auspices (e.g., voluntary nonprofit, proprietary, governmental).

Among short-term hospitals, substantial variation in operating margins is observed across ownership categories. Proprietary–Partnership hospitals (Type 5) exhibit the highest average operating margin (~12.1%), followed by Proprietary–Corporation hospitals (Type 4) at approximately 7.8% and Governmental–Hospital District facilities (Type 11) at roughly 6.2%. Voluntary Nonprofit–Other hospitals (Type 2) average approximately 3.9%, while certain governmental categories—particularly Governmental–City–County (Type 8) and Governmental–County (Type 9)—report negative average margins.

The spread between the highest- and lowest-performing ownership groups exceeds 14 percentage points, indicating meaningful structural heterogeneity within the short-term hospital sector. Proprietary institutions may operate under stronger profit-maximization incentives or cost controls, while governmental hospitals frequently serve safety-net roles and may treat higher proportions of uncompensated or under-reimbursed patients. Additionally, certain public subsidies may not be fully reflected in net patient revenue, potentially affecting reported margins.

Type 11 (Governmental–Hospital District) represents a notable exception among governmental ownership categories, exhibiting average margins comparable to proprietary institutions. This performance may reflect structural differences in governance, such as greater financial autonomy, localized funding mechanisms, or variations in payer composition relative to other governmental facilities. While causal mechanisms cannot be established within this analysis, the result suggests that not all public ownership models operate under the same financial constraints.

Overall, the systematic variation observed across ownership types within short-term hospitals suggests that institutional structure is associated with financial performance and should be incorporated as a categorical control variable in subsequent regression modeling.

EDA Queries — Correlation between **Operating margin** AND **Occupancy rate**, Correlation with Labor Cost Ratio, Correlation with Uncompensated Care

Complete Correlation Summary

Variable	r	Direction	Strength
Occupancy Rate	+0.197	Positive	Weak
Uncompensated Care Ratio	−0.133	Negative	Weak

Labor Cost Ratio	-0.127	Negative	Weak
------------------	--------	----------	------

To evaluate whether selected predictors are meaningfully associated with operating margin, pairwise Pearson correlations were computed.

The correlation between operating margin and occupancy rate is $r = 0.197$, indicating a weak positive relationship. This suggests that hospitals operating at higher utilization levels tend to exhibit modestly higher margins, consistent with economies of scale and fixed cost absorption.

The correlation between operating margin and labor cost ratio is $r = -0.127$, indicating a weak negative relationship. Hospitals allocating a higher proportion of total costs to labor tend to experience slightly lower margins, consistent with labor intensity reducing profitability.

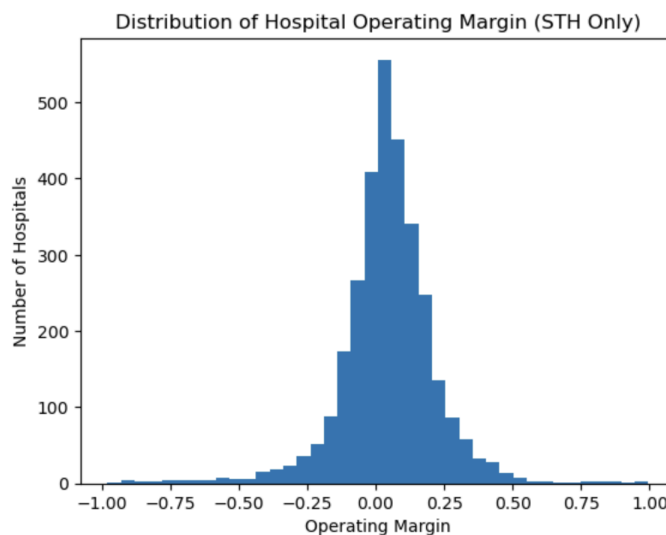
The correlation between operating margin and uncompensated care ratio is $r = -0.133$, also a weak negative relationship. Hospitals with higher shares of uncompensated care appear to face margin pressure, which aligns with financial theory.

Although these correlations are modest in magnitude, their direction is economically intuitive and supports their inclusion in multivariable regression analysis.

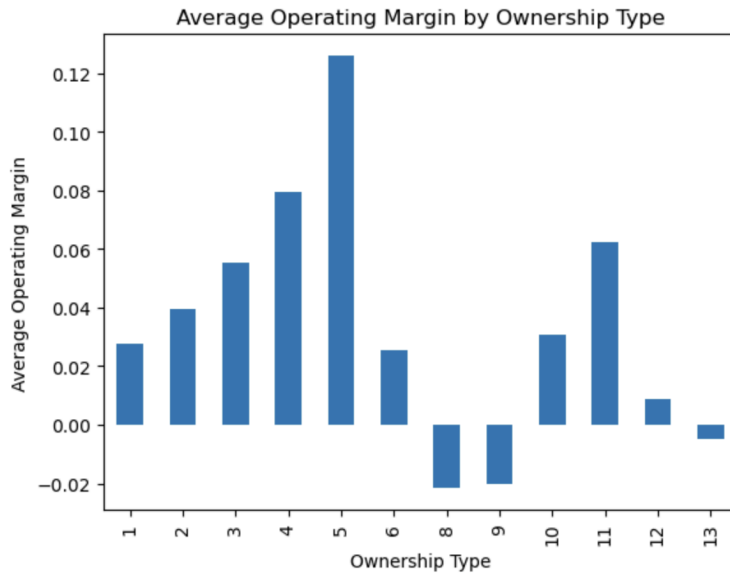
Histogram of Operating Margin

The histogram indicates that operating margin is centered slightly above zero, with most hospitals clustered between -0.25 and $+0.25$. The highest density appears near 0.05 , consistent with the average margin of 4.3% identified in the SQL analysis. Although the distribution exhibits mild right skew — reflecting a small number of high-performing

hospitals — the overall shape is reasonably symmetric without heavy tails or extreme outlier mass that would distort regression estimates. While not perfectly normal, the distribution does not display severe skewness or kurtosis that would invalidate linear modeling assumptions. This supports proceeding with ordinary least squares (OLS) regression to examine conditional relationships between predictors and margin.



Bar Chart: Average Margin by Ownership



The bar chart replicates the SQL findings from Query 2 and highlights substantial heterogeneity across ownership types. For-profit chain hospitals (Type 5) exhibit the highest average operating margin at approximately 12.5%, followed by for-profit independent hospitals (Type 4) at roughly 8% and Type 3 at 5.5%. Non-profit voluntary hospitals (Type 2) average near 4%, while government and hospital district facilities (Types 8 and 9) are the only groups with negative average margins (approximately -1.5% and -2%, respectively). The nearly 15-percentage-point spread between the highest and lowest ownership categories indicates meaningful structural variation in financial performance. These systematic differences suggest that ownership type captures institutional characteristics correlated with profitability and therefore warrants inclusion as a categorical control in the regression model.

After everything, I chose four hypotheses.

1. If occupancy increases, then operating margin increases.
2. If labor cost ratio increases, then operating margin decreases.
3. If uncompensated care ratio increases, then operating margin decreases.
4. If ownership structure matters, then margins differ across ownership types holding the other variables constant.

The variables once again are operating_margin as the dependent variable and occupancy_rate, labor_cost_ratio, uncompensated_ratio, and type_of_control as the independent variables

Model 1 - Bivariate OLS

To establish a starting point and test H1, I estimated a simple regression of operating margin on occupancy rate. Occupancy has a positive and statistically significant relationship with operating margin ($\beta = 0.1466$, $p < 0.001$). A 10 percentage-point increase in occupancy is associated with approximately a 1.5 percentage-point increase in operating margin. This is intuitive: hospitals with more beds filled generate more revenue while fixed costs remain largely unchanged.

However, the model explains only 3.9% of the variation in operating margins ($R^2 = 0.039$). Occupancy alone does not explain why hospital margins differ so widely. This suggests that cost structure and other financial factors must also contribute to profitability differences. These limitations motivate the transition to a multivariate model incorporating labor costs and uncompensated care.

Relevant code snippet:

```
from scipy import stats
clean_df = df[['occupancy_rate', 'operating_margin']]
clean_df = clean_df.replace([np.inf, -np.inf], np.nan)
clean_df = clean_df.dropna()

x = clean_df['occupancy_rate']
y = clean_df['operating_margin']

slope, intercept, r_value, p_value, st_err = stats.linregress(x, y)
```

Model 2: Multivariate OLS

To jointly evaluate H1–H3, I estimated a multivariate linear regression in which operating margin was modeled as a function of occupancy rate, labor cost ratio, and uncompensated care ratio. The specification allows each predictor to be interpreted holding the others constant, isolating their independent associations with financial performance.

The model explains approximately 7.0% of the cross-sectional variation in operating margins ($R^2 = 0.070$; Adjusted $R^2 = 0.069$). While modest in absolute terms, this represents a substantial improvement over the bivariate occupancy model ($R^2 = 0.039$), nearly doubling explanatory power. The F-statistic is highly significant ($p < 0.001$), indicating that the predictors are jointly informative. All three coefficients are statistically significant at conventional levels ($p < 0.001$) and display economically meaningful magnitudes.

Occupancy rate exhibits a positive association with operating margin ($\beta = 0.1409$). Interpreted economically, a 10 percentage-point increase in occupancy is associated with approximately a 1.4 percentage-point increase in margin, holding cost structure and uncompensated care constant. This finding supports H1 and is consistent with the intuition that

higher capacity utilization improves financial performance through better absorption of fixed costs.

Labor cost ratio is negatively associated with margin ($\beta = -0.1816$). A 10 percentage-point increase in labor cost share corresponds to roughly a 1.8 percentage-point decline in operating margin. The magnitude of this effect exceeds that of occupancy, suggesting that cost discipline plays a more powerful role in profitability than utilization alone. This provides strong support for H2.

Uncompensated care ratio has the largest coefficient in absolute value ($\beta = -0.3048$). A 10 percentage-point increase in uncompensated care burden is associated with a 3.0 percentage-point reduction in operating margin. This result supports H3 and indicates that reimbursement exposure represents the strongest structural constraint on hospital profitability among the included predictors.

Despite statistical significance and stable coefficient signs, overall explanatory power remains limited. The R^2 of 0.07 indicates that the majority of variation in hospital margins is driven by additional institutional and market-level factors not captured in this specification. The model therefore provides clear evidence of systematic directional effects. Utilization improves margins, labor costs compress them, and uncompensated care exerts the strongest negative pressure — but it does not fully account for the variability in hospital financial outcomes.

Residual Diagnostics

The diagnostic statistics show clear non-normality in the residuals. Kurtosis is 8.694, and the Jarque–Bera test strongly rejects normality ($p \approx 0$), indicating heavy tails. This means extreme margin values occur more often than would be expected under a normal distribution.

At the same time, the residual plot does not show any obvious curvature or nonlinear pattern. Residuals are centered around zero, suggesting the model is not systematically over- or under-predicting operating margins. This supports the use of a linear specification.

Although the residual distribution is heavy-tailed, the large sample size (over 3,000 observations) helps ensure that OLS coefficient estimates remain reliable under large-sample theory. Still, the outliers

OLS Regression Results						
Dep. Variable:	operating_margin	R-squared:	0.070			
Model:	OLS	Adj. R-squared:	0.069			
Method:	Least Squares	F-statistic:	75.86			
Date:	Sun, 22 Feb 2026	Prob (F-statistic):	2.63e-47			
Time:	01:41:25	Log-Likelihood:	1099.6			
No. Observations:	3033	AIC:	-2191.			
Df Residuals:	3029	BIC:	-2167.			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0

const	0.0705	0.014	5.074	0.000	0.043	
0.098						
occupancy_rate	0.1409	0.013	10.780	0.000	0.115	
0.167						
labor_cost_ratio	-0.1816	0.026	-7.061	0.000	-0.232	-
0.131						
uncompensated_ratio	-0.3048	0.045	-6.761	0.000	-0.393	-
0.216						
=====						
Omnibus:	443.707	Durbin-Watson:	1.834			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	4170.307			
Skew:	-0.381	Prob(JB):	0.00			
Kurtosis:	8.694	Cond. No.	18.1			

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

and heavy tails motivated testing Ridge regression as a stability check.

Relevant code:

```
import statsmodels.api as sm
X = df[['occupancy_rate', 'labor_cost_ratio', 'uncompensated_ratio']].dropna()
y = df.loc[X.index, 'operating_margin']

X1 = sm.add_constant(X)

model = sm.OLS(y, X1)
results = model.fit()
print(results.summary())
```

Residual Plot (OLS)

The residual plot shows that residuals are centered around zero, which suggests the model is not systematically over- or under-predicting operating margins. There is no obvious curvature in the plot, so a linear specification appears reasonable.

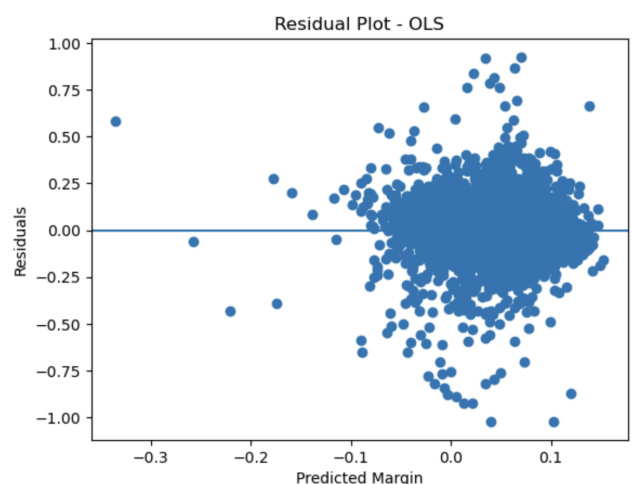
That said, the spread of the residuals is not perfectly even. There are several extreme values, and the dispersion increases slightly at higher predicted margins. This aligns with the earlier diagnostic statistics showing heavy tails. In other words, extreme margin values occur more often than a normal distribution would predict.

Because the sample size is large (around 3,000 observations), the OLS estimates remain reliable. Still, the presence of outliers and uneven variance made it worth testing Ridge regression as a stability check.

Code snippet :

```
predictions = results.predict(X1)
residuals = y - predictions
```

```
plt.scatter(predictions, residuals)
plt.axhline(0)
plt.xlabel("Predicted Margin")
plt.ylabel("Residuals")
plt.title("Residual Plot - OLS")
plt.show()
```



Model Stability and Regularization (Ridge)

Because the residual diagnostics showed heavy tails and some extreme observations, I tested Ridge regression to see whether shrinking the coefficients would improve stability or predictive performance. Ridge adds a penalty term that reduces coefficient magnitude, which can help when models are sensitive to outliers or multicollinearity.

I implemented Ridge regression using cross-validation to select the optimal regularization parameter (α). The selected α was relatively large (≈ 104.81), indicating strong penalization. In terms of fit, Ridge performed almost identically to OLS. The in-sample R^2 was 0.0698, compared to 0.070 for OLS. Although the coefficients were substantially shrunk (Occupancy: 0.032; Labor cost ratio: -0.021 ; Uncompensated ratio: -0.020), this shrinkage did not improve explanatory power. In short, regularization reduced coefficient magnitude but did not meaningfully change model performance, suggesting that the original OLS estimates were already reasonably stable.

Train/Test Model Comparison

To evaluate how well the models generalize, I split the data into training and test sets (80/20). Both OLS and Ridge were estimated on the training set, and performance was evaluated on the held-out test set using R^2 and RMSE.

Out of sample, OLS slightly outperformed Ridge. The OLS model achieved a test R^2 of 0.0589 and an RMSE of 0.1648, compared to Ridge's test R^2 of 0.0407 and RMSE of 0.1664.

The differences are not dramatic, but shrinkage did not improve predictive performance. This suggests that multicollinearity is not a major concern in this dataset and that the OLS coefficients are already reasonably stable. Ridge reduced coefficient magnitudes, but that reduction did not translate into better generalization. Overall, the simpler OLS specification remains preferable for this analysis.

Predictive Performance Visualization #2

The scatter plot compares actual operating margins to predicted margins from the OLS model on the test set. The 45-degree line represents perfect prediction. While the points generally follow the upward direction of the line, there is noticeable dispersion around it.

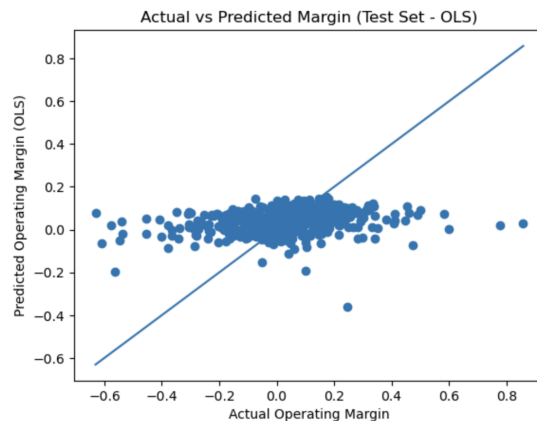
This pattern is consistent with the modest test R^2 (≈ 0.059). The model captures the overall directional relationship between predictors and margins, but it does not precisely predict individual outcomes. Errors become larger for very high and very low margins, indicating that extreme financial outcomes are not fully explained by the included variables.

Overall, the model identifies systematic trends but has limited precision in predicting hospital-level margins.

Code Snippet

```
import matplotlib.pyplot as plt

plt.figure()
plt.scatter(y_test, ols_preds)
plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()])
plt.xlabel("Actual Operating Margin")
plt.ylabel("Predicted Operating Margin (OLS)")
plt.title("Actual vs Predicted Margin (Test Set)")
plt.show()
```



Final Reflection on Modeling Choices

This modeling process progressed from a simple bivariate specification to a multivariate framework that incorporated financial structure and ownership, followed by diagnostic evaluation, regularization testing, and out-of-sample validation. The objective was not only to estimate coefficients, but to assess stability and generalizability.

Explanatory power increases meaningfully across specifications, rising from 3.9% in the bivariate model to 9.8% in the fully specified model with ownership controls. This improvement indicates that governance structure contributes independent explanatory power beyond occupancy, labor intensity, and uncompensated care burden. However, overall predictive strength remains modest. Even the most complete specification explains less than 10% of variation in operating margins.

Ridge regression did not improve out-of-sample performance, suggesting that the OLS specification is already reasonably stable and that multicollinearity is not a primary concern. Adding regularization reduced coefficient magnitudes but did not improve generalization, so the simpler OLS model remains appropriate.

The limited R^2 highlights an important substantive point rather than a modeling flaw. Hospital operating margins are shaped by structural forces not fully captured in this dataset, such as regional market power, payer mix composition, competitive intensity, management quality, and regulatory constraints. The model identifies statistically meaningful directional relationships, but margins remain inherently difficult to predict in a linear framework. The results therefore provide explanatory insight into key drivers of profitability, while acknowledging the broader institutional complexity of hospital financial performance.

Hypothesis 4: Does Ownership Type Add Predictive Power?

To test whether ownership structure adds predictive power beyond financial variables, I extended the baseline regression by including dummy variables for ownership type. Because ownership is categorical, it cannot be entered directly into an OLS model. I therefore converted the variable into indicator variables using `pd.get_dummies(drop_first=True)`, omitting one category as the reference group to avoid perfect multicollinearity. This allows the ownership coefficients to be interpreted relative to that baseline category while controlling for occupancy, labor cost ratio, and uncompensated care ratio.

The baseline financial model explains approximately 7.0% of the variation in operating margins ($R^2 \approx 0.070$). After adding ownership controls, explanatory power increases to 9.8% ($R^2 = 0.098$; Adjusted $R^2 = 0.094$). This represents a 2.8 percentage-point increase in explained variance. Although overall R^2 remains modest, the increase is meaningful in a financial dataset where margins are influenced by many unobserved structural factors. These results support Hypothesis 4: ownership structure contributes additional explanatory power beyond cost structure and utilization.

The extended model is statistically significant overall (F-statistic = 23.95, $p < 0.001$), indicating that the combined set of financial and ownership variables explains systematic differences in operating margins. Importantly, the original financial predictors remain statistically significant and stable in magnitude after adding ownership controls. Occupancy remains positively associated with margins (0.1707), while labor cost ratio (−0.1483) and uncompensated care ratio (−0.3029) remain negative and economically substantial. This indicates that utilization, labor intensity, and reimbursement burden are robust drivers of profitability, independent of governance structure.

Several ownership categories are statistically significant relative to the reference group, suggesting that certain governance structures are associated with systematically higher margins even after controlling for operational and financial variables. Other ownership categories are not statistically significant, indicating heterogeneity across institutional models rather than a uniform ownership effect.

Residual diagnostics are consistent with earlier specifications. The residual distribution remains heavy-tailed (kurtosis ≈ 8.75 ; Jarque–Bera $p \approx 0$), but there is no indication of severe multicollinearity (Condition Number ≈ 28.6). With approximately 3,100 observations, inference remains reliable under large-sample theory.

Overall, hospital operating margins appear to reflect three distinct influences: operational efficiency, financial burden, and governance structure. Ownership increases explanatory power from roughly 7% to nearly 10%, indicating that institutional design plays a measurable role in financial performance alongside cost discipline and utilization.

Relevant Code:

```
import pandas as pd
import statsmodels.api as sm
```

```
# Create dummy variables
ownership_dummies = pd.get_dummies(df['type_of_control'], drop_first=True)
```

```
# Combine with previous predictors
X_extended = pd.concat([
    df[['occupancy_rate', 'labor_cost_ratio', 'uncompensated_ratio']],
    ownership_dummies
], axis=1)
```

```
X_extended = sm.add_constant(X_extended)
y = df['operating_margin']
```

```
model_extended = sm.OLS(y,
X_extended).fit()
```

```
print(model_extended.summary())
```

OLS Regression Results						
Dep. Variable:	y	R-squared:	0.098			
Model:	OLS	Adj. R-squared:	0.094			
Method:	Least Squares	F-statistic:	23.95			
Date:	Sun, 22 Feb 2026	Prob (F-statistic):	1.36e-59			
Time:	02:25:35	Log-Likelihood:	1152.8			
No. Observations:	3103	AIC:	-2276.			
Df Residuals:	3088	BIC:	-2185.			
Df Model:	14					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0124	0.016	0.768	0.443	-0.019	0.044
x1	0.1707	0.013	12.768	0.000	0.144	0.197
x2	-0.1483	0.026	-5.600	0.000	-0.200	-0.096
x3	-0.3029	0.046	-6.568	0.000	-0.393	-0.212
x4	0.0145	0.009	1.575	0.115	-0.004	0.033
x5	0.0360	0.064	0.564	0.573	-0.089	0.161
x6	0.0697	0.011	6.478	0.000	0.049	0.091
x7	0.1183	0.021	5.616	0.000	0.077	0.160
x8	0.0252	0.026	0.978	0.328	-0.025	0.076
x9	0.0007	0.026	0.027	0.978	-0.051	0.053
x10	0.0009	0.017	0.051	0.959	-0.033	0.035
x11	-0.0102	0.025	-0.415	0.678	-0.059	0.038
x12	0.0772	0.018	4.400	0.000	0.043	0.112
x13	0.0406	0.031	1.298	0.194	-0.021	0.102
x14	-0.0230	0.026	-0.892	0.373	-0.073	0.028
Omnibus:	490.445	Durbin-Watson:	1.875			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	4384.357			
Skew:	-0.472	Prob(JB):	0.00			
Kurtosis:	8.746	Cond. No.	28.6			

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.